NETWORKING PRACTICAL ASSIGNMENT – 2

Static IP • DHCP • DNS • VLAN • NAT

Student Name: _____ **Date:** _____

Objective: Practise five important network configuration tasks using a home lab or Cisco Packet Tracer. The IP addresses below are examples; adjust them to match your lab.

Sample Network Plan

Purpose	Example setting
LAN subnet	192.168.10.0/24
Router / default gateway	192.168.10.1
Static device IPs	192.168.10.11 to 192.168.10.15
DHCP pool	192.168.10.100 to 192.168.10.150
Example local DNS/web server	192.168.10.50
VLAN 10 / VLAN 20	Separate VLANs; use separate subnets for routing labs

1. Assign Static IP Addresses to Five Devices

Aim: Give five computers unique static IP addresses in the same subnet and verify communication.

Steps:
1. Connect five laptops/desktops to the same router or switch.
2. On each Windows device, open Settings → Network & Internet → Advanced network settings → More network adapter options.
3. Right-click the active adapter → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties.
4. Select “Use the following IP address” and enter the values below.

Device	IP address	Subnet mask	Default gateway
PC-1	192.168.10.11	255.255.255.0	192.168.10.1
PC-2	192.168.10.12	255.255.255.0	192.168.10.1
PC-3	192.168.10.13	255.255.255.0	192.168.10.1
PC-4	192.168.10.14	255.255.255.0	192.168.10.1
PC-5	192.168.10.15	255.255.255.0	192.168.10.1

Verify: Open Command Prompt and run `ipconfig`, then test from PC-1:
`ping 192.168.10.12`
`ping 192.168.10.13`
Repeat tests between the other devices.

Expected result: All five devices have different addresses in the same subnet and can ping one another, if cabling, Wi-Fi, and firewall settings permit ICMP.

2. Configure DHCP for at Least Three Devices

Aim: Automatically assign unique IP addresses to three or more clients.

Using a home router:
1. Sign in to the router's admin page.
2. Find LAN / DHCP Server settings and enable DHCP.
3. Set the address pool to **192.168.10.100–192.168.10.150**.
4. Set the gateway to **192.168.10.1**; set DNS to the router or lab DNS server.
5. Save. On at least three clients, select “Obtain an IP address automatically”.

Packet Tracer router example: Configure the router LAN interface as 192.168.10.1/24 and enable it. Then enter:

```
enable
configure terminal
```

```
ip dhcp excluded-address 192.168.10.1 192.168.10.99
ip dhcp pool LAB-POOL
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 192.168.10.2
exit
```

On each Packet Tracer PC, choose Desktop → IP Configuration → DHCP. On Windows, run `ipconfig /all`. Confirm that at least three clients have different addresses from the pool, the correct subnet mask, and gateway.

Expected result: Each DHCP client receives a unique lease from the configured range.

3. Configure a Local DNS Hostname

Aim: Make a local web page or device interface open using a name such as musicplayer.local.

Quick test using the Windows hosts file:
1. Run a local web server on a lab PC and note its IP, for example **192.168.10.50**. First test that `http://192.168.10.50` opens.
2. Open Notepad as Administrator.
3. Open `C:\Windows\System32\drivers\etc\hosts`. Choose “All files” if the file is not visible.
4. Add this line and save:

```
192.168.10.50 musicplayer.local
```

5. In the browser, visit `http://musicplayer.local`. For the hosts-file method, add the same entry on every client that needs the name.

Central DNS alternative: Configure dnsmasq or your router's local DNS with an A record: `musicplayer.local` → `192.168.10.50`. Set clients to use that DNS server. Test with `nslookup musicplayer.local`.

Expected result: The hostname resolves to the local server and its page opens. The sample IP is only an example.

4. Create Two VLANs and Test Communication

Aim: Create VLAN 10 (Workstations) and VLAN 20 (Smart Devices). Same-VLAN devices should communicate. Different VLANs should remain isolated when inter-VLAN routing is not configured.

Example switch ports: PC1 and PC2 on Fa0/1–Fa0/2 (VLAN 10); Device3 and Device4 on Fa0/3–Fa0/4 (VLAN 20).

Cisco switch CLI:

```
enable
configure terminal
vlan 10
name Workstations
exit
vlan 20
name Smart-Devices
exit
interface range fa0/1-2
switchport mode access
switchport access vlan 10
exit
interface range fa0/3-4
switchport mode access
switchport access vlan 20
exit
end
show vlan brief
```

Assign example IPs:
PC1: 192.168.10.11/24
PC2: 192.168.10.12/24
Device3: 192.168.20.11/24
Device4: 192.168.20.12/24
For this basic isolation test, leave default gateways blank.

Ping tests:
• From PC1: `ping 192.168.10.12` — should succeed.
• From Device3: `ping 192.168.20.12` — should succeed.
• From PC1: `ping 192.168.20.11` — should fail because the devices are in different VLANs and no router/L3 switch is routing between them.

Important: VLANs are separate Layer-2 broadcast domains. To allow communication between VLANs, configure inter-VLAN routing. Do not configure it for this basic isolation demonstration.

5. Configure NAT to Simulate Internet Access

Aim: Translate private LAN addresses to an outside address so LAN clients can reach an external or simulated server.

Example topology: LAN PC → Router G0/0 (inside) → Router G0/1 (outside) → simulated ISP / external server. Example LAN: 192.168.10.0/24; inside router: 192.168.10.1; outside interface: 203.0.113.2/30; ISP next hop: 203.0.113.1. These public documentation addresses are for simulation only.

Router NAT example: Change interface names and outside addressing to match your topology.

```
enable
configure terminal
interface g0/0
ip address 192.168.10.1 255.255.255.0
ip nat inside
no shutdown
exit
interface g0/1
ip address 203.0.113.2 255.255.255.252
ip nat outside
no shutdown
exit
access-list 1 permit 192.168.10.0 0.0.0.255
ip nat inside source list 1 interface g0/1 overload
ip route 0.0.0.0 0.0.0.0 203.0.113.1
end
```

Test:
1. Configure the LAN PC as 192.168.10.11/24, gateway 192.168.10.1.
2. Ping the gateway: `ping 192.168.10.1`.
3. Ping the simulated external server's IP address.
4. If DNS is configured, try a hostname such as www.wikipedia.org. A simulator may require you to create an external server

and DNS record; public websites are not guaranteed to work in Packet Tracer.

Verify NAT: Generate traffic, then on the router run `show ip nat translations` and `show ip nat statistics`.

Expected result: LAN traffic is translated to the router's outside interface address and reaches the simulated server when routing and server settings are correct.

Final Verification Checklist

Task	What to confirm	Done
Static IP	Five unique IPs in one subnet; ping works.	■
DHCP	At least three clients get unique leases from the pool.	■
DNS	musicplayer.local resolves and opens the local page.	■
VLAN	Same-VLAN ping succeeds; cross-VLAN ping fails without routing.	■
NAT	LAN reaches simulated outside server; NAT translation is visible.	■

Conclusion: In this practical, I configured static IP addressing, DHCP, local hostname resolution, VLAN segmentation, and NAT. I used ping and network commands to verify the results.

Safety note: Perform these exercises only on your own home/lab network or an authorized simulator. Record your original router settings before making changes.